



JTAG (IEEE 1149.1/P1149.4) Tutorial Introductory

Based on TI Test Symposium (1997)

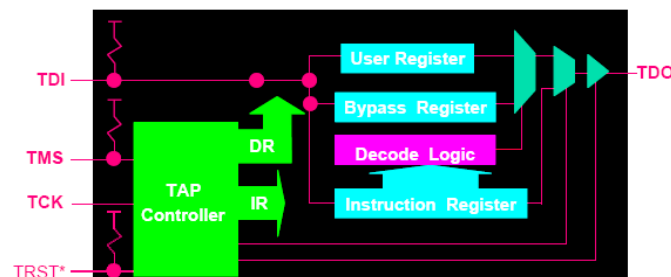


Standard Approach To Test



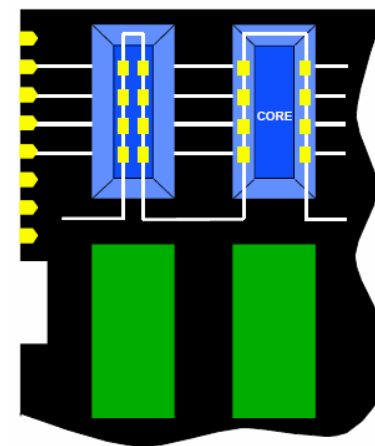
- Developed by Joint Test Action Group (over 200 SC, test, and system vendors) starting in mid '80's
- Sanctioned by IEEE as Std 1149.1 Test Access Port and Boundary-Scan Architecture in 1990
- Solution: Build test facilities/test points into chips
- Focus: Ensure compatibility between all compliant ICs

Standard Test Access Port (TAP)



- 4/5-Wire Interface at Chip-Level
- Serial Instruction/Serial Data Port
- Extensible to Include
 - user-defined instructions
 - user-defined data registers

Basic Boundary-Scan Architecture

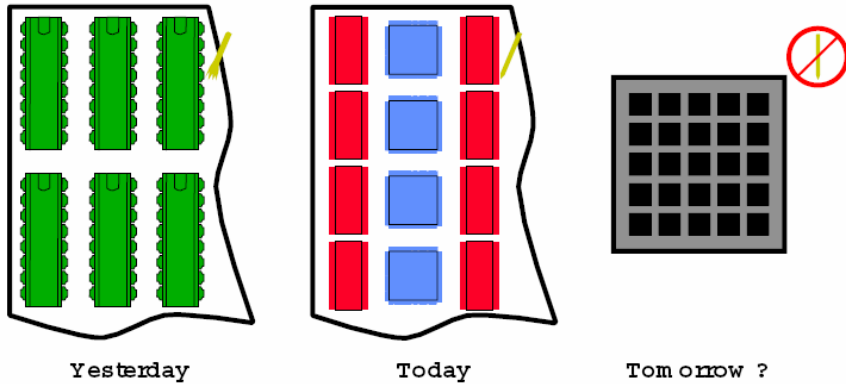


- Scan effectively partitions digital logic to facilitate control and observation of its function
- Chip-Internal Scan: Partitions chips at storage cells (latches/ flip-flops) to effectively partition sequential logic into clusters of combinational logic
- Boundary-Scan: Partitions boards at chip I/Os for control and observation of board-level nodes

Why is JTAG necessary?



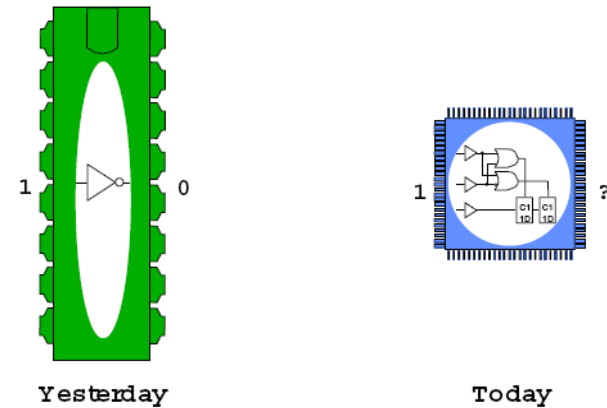
- Miniaturization results in loss of test access



Ever-Expanding Chip



- Increasing integration at chip level complicates controllability



Can't Afford Not To Test



Cost will increase by a factor of ten as fault finding moves from one level of complexity to the next. The result:

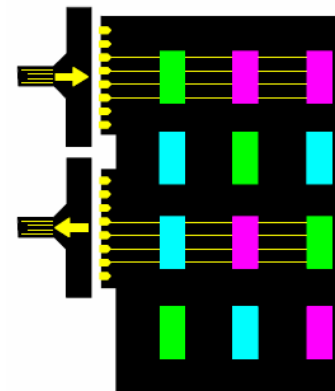
- Reduced Profit Margins
- Delayed Product Introduction
- Dissatisfied Customers

1. Device level	1 unit of cost
2. Board level	10 units of cost
3. System level	100 units of cost
4. Field level	1,000 units of cost

Conventional Methods of Board Test (in 90's)



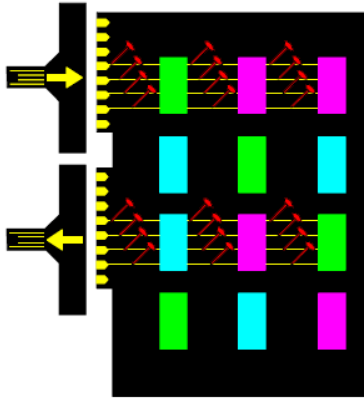
Functional Test (‘Edge-Connector’ Test)



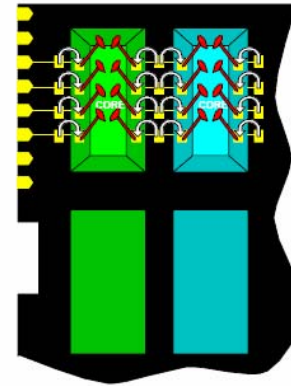
- Based on board function, rather than structure
- Test generation primarily manual
- Test access limited to primary I/O only



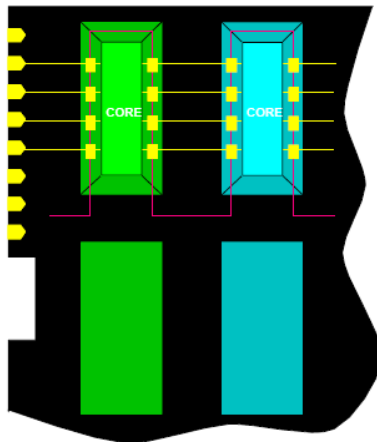
In-Circuit Test (*'Bed-of-Nails' Test*)



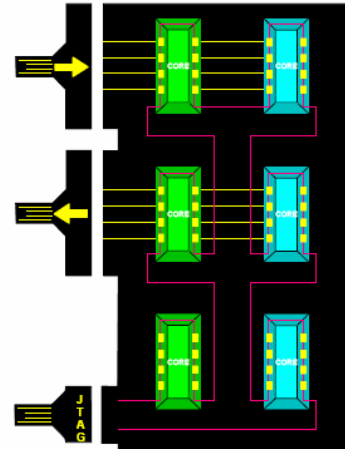
- Based on board structure, but limited by chip complexity
- Expensive testers and fixtures required
- Test access limited by:
 - Fine pitch packages
 - Double-sided boards
 - Conformal coating
 - MCMs



- 'In-Circuit' test points moved onto the silicon, creating 'Virtual Nails'
- Boundary scan cells bound each net, providing for continuity testing
- Observe/Control cells provide for test and normal function

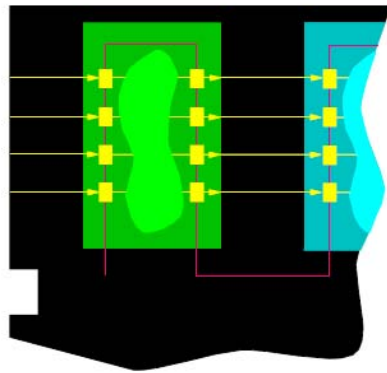


- Scan provides a means to arbitrarily observe test results and source test stimulus
- Scan method requires minimal on chip/board resources (pins/nets)



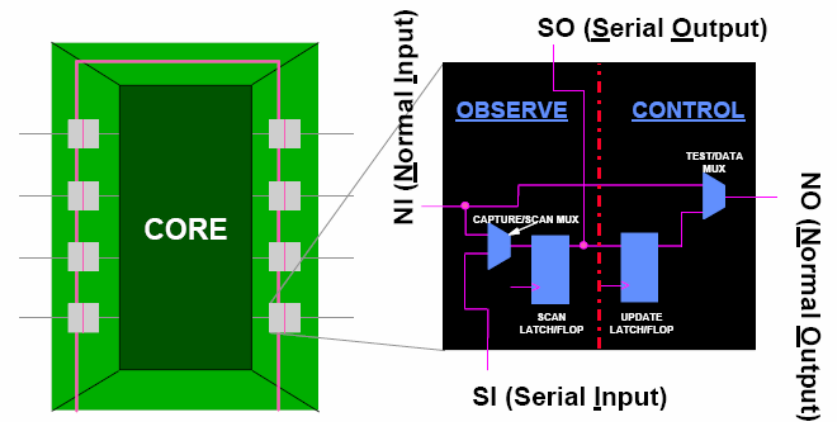
- Based on board structure; Not limited by chip function/ complexity
- Test access is not limited by board physical factors

Boundary Scan for Board Test (con't)

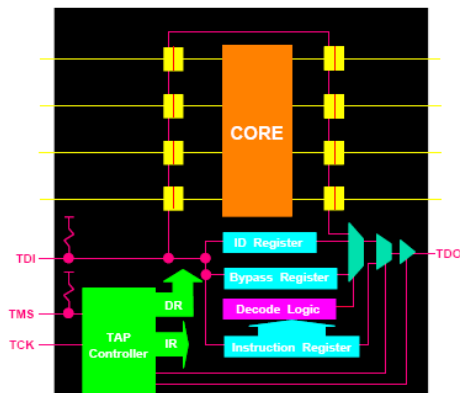


- Chip function need not be considered for board test (shorted/open nets)
- Test generation is highly automated; Simple 'In-Circuit Library' models (BSDL) are vendor-supplied or EDA-generated

Boundary Scan Cell

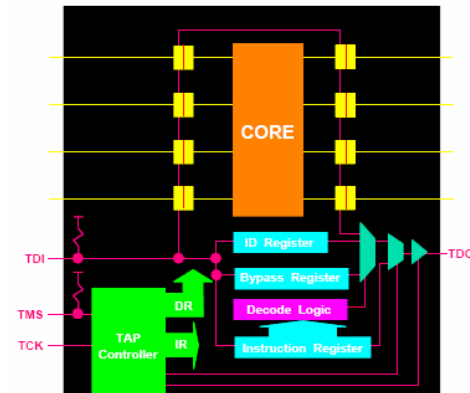


The Control Architecture



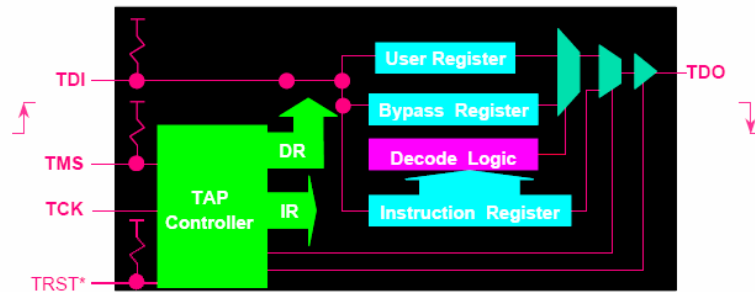
- Boundary scan and other test data registers operate under control of instruction register
- Data is scanned from TDI to TDO through selected test data register or instruction register under control of Test Access Port (TAP) controller
- TAP operates synchronously to TCK using TMS for state selection

The Test Logic Architecture



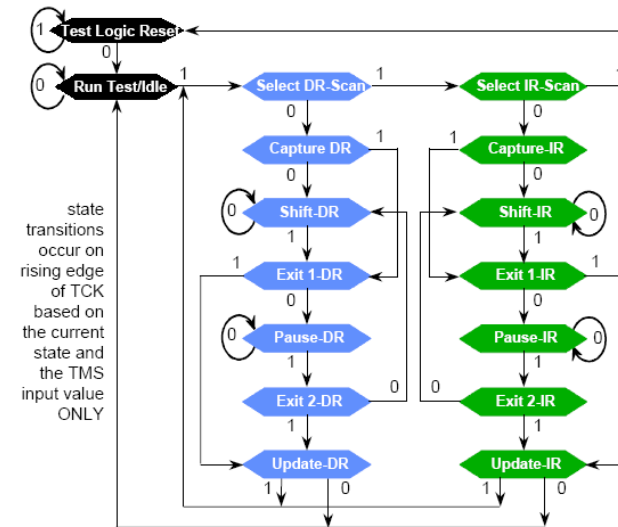
- The REQUIRED elements of the test logic architecture are:
 - Test Access Port (TAP)
 - TAP Controller
 - One Instruction Register
 - A Set of Data Registers
- All such elements are REQUIRED to be dedicated test logic (not used for any other purpose) with the exception of user-defined data registers

The Test Access Port



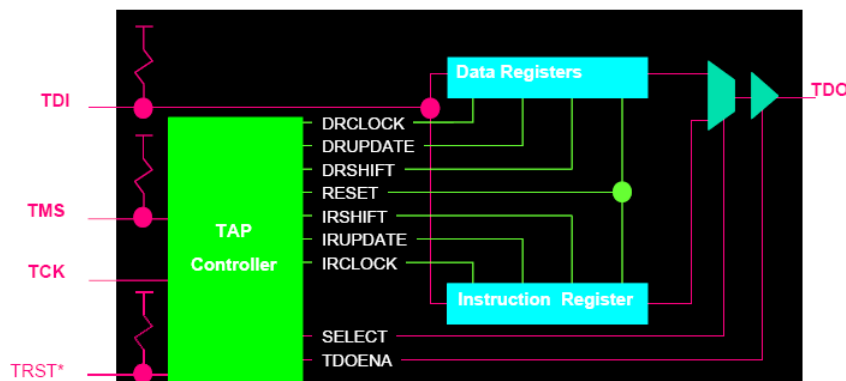
- 4 wire TAP interface REQUIRED
- either power-up reset or 5th wire, TRST*, is REQUIRED
- all TAP pins are REQUIRED to be dedicated (not used for any other purpose)
- pullups REQUIRED at TDI and TMS (also, TRST*, if implemented)
- all inputs (e.g., TDI) are REQUIRED to be sampled on TCK rising, all outputs (e.g. TDO) are REQUIRED to be propagated on TCK falling

The TAP Controller



- The TAP controller is REQUIRED to conform exactly to this state diagram
- The instruction scan sequence is REQUIRED to access only the single instruction register
- The data scan sequence is REQUIRED to access only the data register selected by the current instruction
- TDO is REQUIRED to be active only in the Shift-IR and Shift-DR states

The TAP Controller Outputs



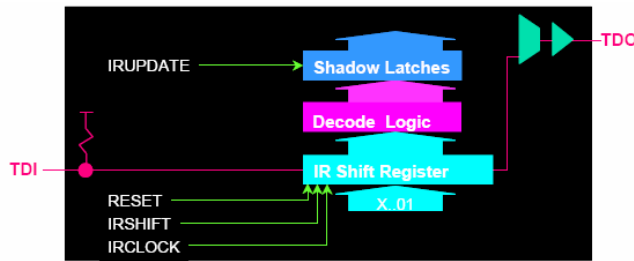
- IR scan and update clocks and shift/capture select
- DR scan and update clocks and shift/capture select
- TDO enable and TDO instruction/data select

Register Summary



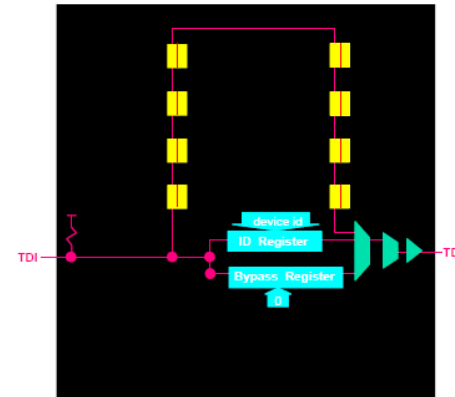
Register (REQUIRED/Optional)	Length	Capture Value	Selected by Instruction(s)
<u>INSTRUCTION</u>	at least two	X..01	n/a
<u>BOUNDARY</u>	user-specified	user-specified	Extest Sample/Preload Intest Clamp HighZ
<u>BYPASS</u>	exactly one	0	Bypass
Device ID	exactly 32	X..1	Idcode Usercode
user-defined	user-specified	user-specified	user-specified

The Instruction Register



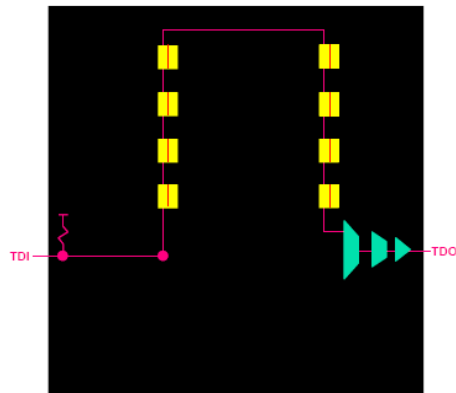
- The instruction register is REQUIRED to be at least 2 bits in length
- The instruction register is REQUIRED to capture the binary value 01 in its 2 least-significant bits on the rising edge of TCK in Capture-IR (other bits may capture user-specified data)
- The instruction register is REQUIRED to have a latched parallel output such that a new instruction only takes effect after shifting is complete (on falling edge of TCK in Update-IR)
- The instruction register is REQUIRED to reset to Idcode (if implemented), otherwise to Bypass

The Test Data Registers



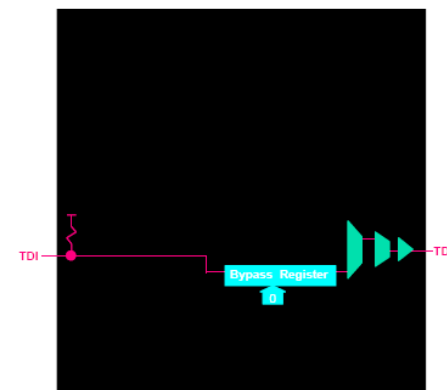
- Only 2 test data registers are REQUIRED:
 - The boundary-scan register - the serial concatenation of all boundary-scan cells, at least one for each digital signal pin
 - The bypass register
- These registers, and the device identification register, if implemented, are REQUIRED to be dedicated test logic (used for no other purpose)

The Boundary-Scan Register



- The boundary-scan register is REQUIRED to be a concatenation of all boundary-scan cells
 - it is REQUIRED that there be at least one BSC for each digital input or output to the system logic (including on-chip outputs to analog circuitry)
 - it is REQUIRED that there NOT be BSCs at
 - » TAP pins
 - » compliance enables
 - » non-digital signals

The Bypass Register



- The bypass register is REQUIRED to be one bit in length
- The bypass register is REQUIRED to capture a logic 0 value in Capture-DR state
- It is REQUIRED that any operation of the bypass register have no effect on the operation of the system logic

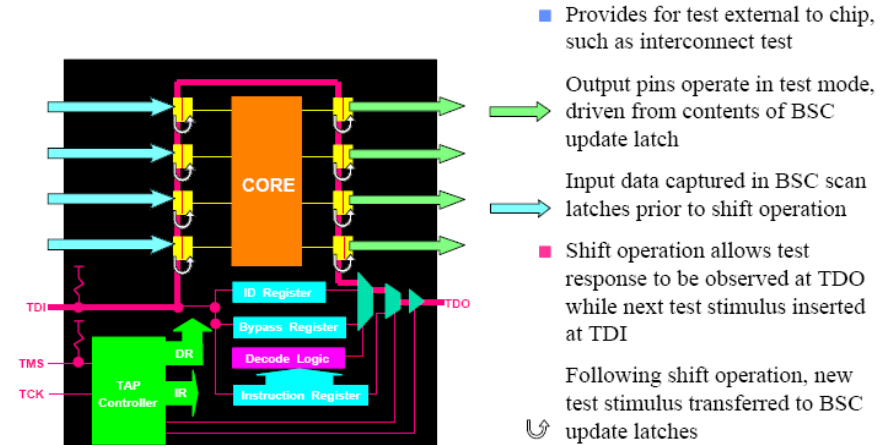
Instruction Summary



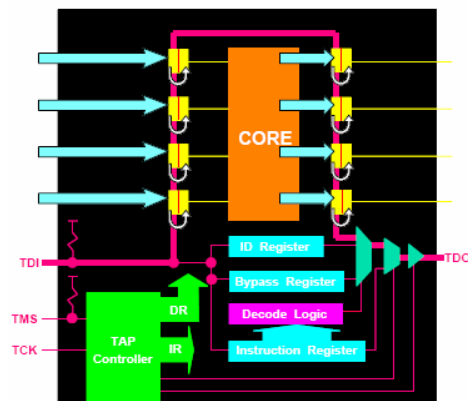
Instruction (REQUIRED/ Optional)	Opcode	Mode	Selected Data Register
EXTEST	0..0*	Test	Boundary
SAMPLE/PRELOAD	user-specified	Normal	Boundary
BYPASS	1..1	Normal	Bypass
Intest	user-specified	Test	Boundary
Runbist	user-specified	Test	User-specified
Idcode	user-specified	Normal	Device ID
Usercode	user-specified	Normal	Device ID
Clamp	user-specified	Test	Bypass
HighZ	user-specified	Test	Bypass
user-defined	user-specified	user-specified	user-specified

* - 1148.1 revision in progress will remove the requirement for EXTEST opcode of all zero

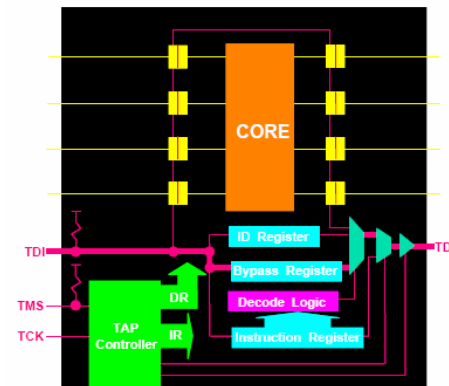
The Extest Instruction



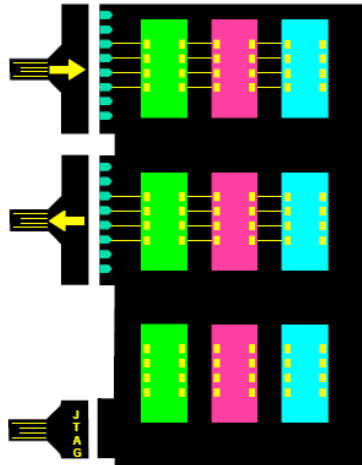
The Sample/Preload Instruction



The Bypass Instruction



Interconnect Test with JTAG Board

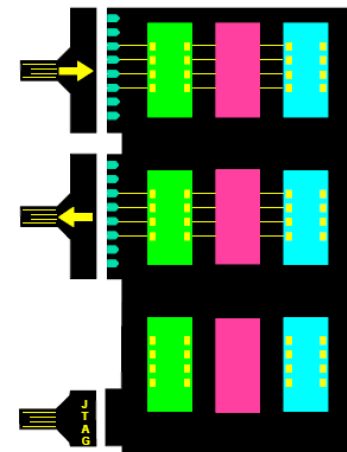


- All nets bound by BSC's and/or primary I/O requiring no physical access
- Parallel access reduced to card edge only
- Test generation and application fast and easy

Partial Boundary Scan Board Test

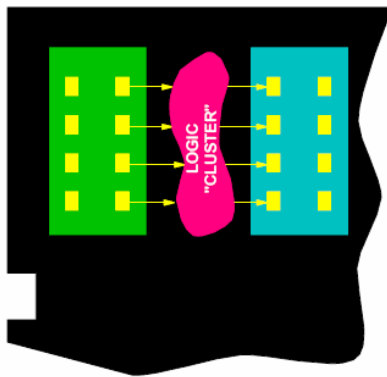


Partial B/S Board



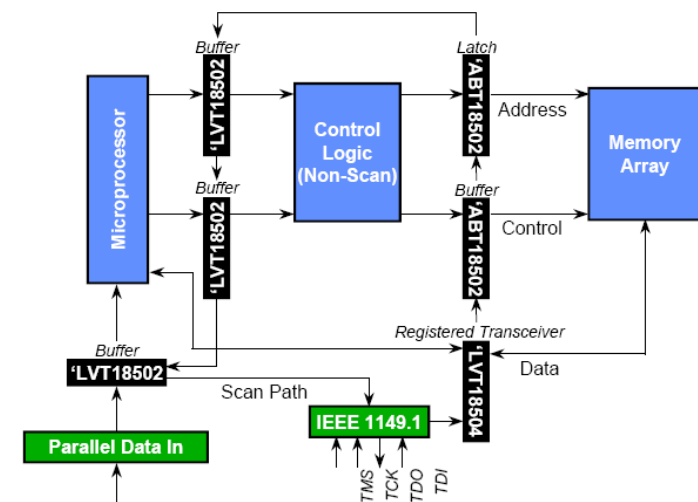
- Not all nets are bound by boundary scan and/or primary I/O, perhaps requiring some ICT access
- Expense and complexity reduced for test generation and test application for chips/nets with B/S access
- Cluster testing may be used to access non-scan nets

Glue Logic Test

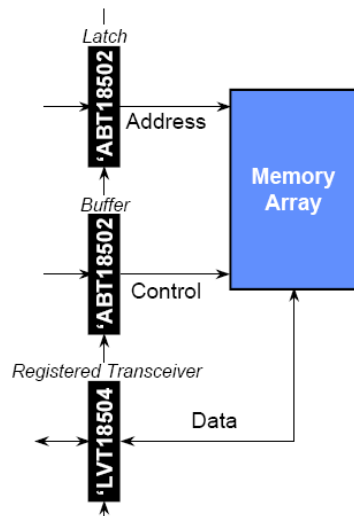


- Random-logic cluster is bound by boundary-scannable chips
- Deterministic test stimulus (ATPG-generated) can be driven to cluster from B/S outputs
- Test response can be captured at B/S inputs
- BIST methods (PRPG/PSA) can be used for increased test throughput and near "At-speed" performance

Logic Cluster Test

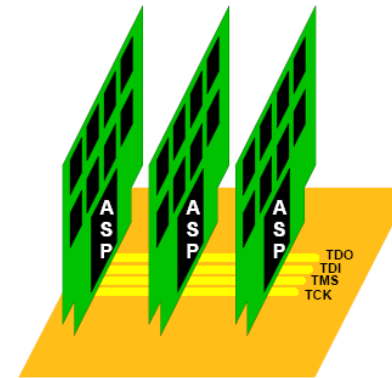


Memory Test



- Memory array bound by boundary scan chips
- Automatic test patterns can be generated and driven from B/S outputs
- Test response can be captured at B/S inputs
- Transceivers can test for net shorts w/o memory R/W
- BIST methods (PRPG/PSA) can be used for increased test throughput

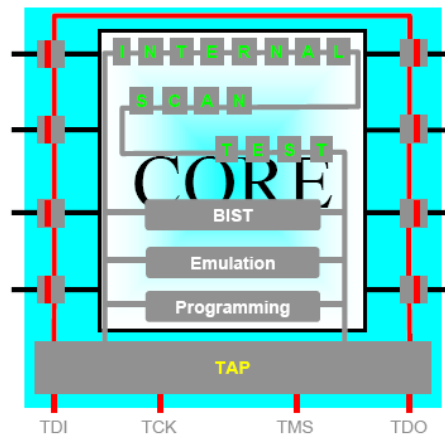
System-Level Test



- TAP-addressable interface unit extends JTAG access beyond board-level
- System-level test
- System design verification
- Sys integration (Mfg test)
- Sys self-test (Field Svc)
- Supports in place board test and board-to-board test
- Allows reuse of device/board test data

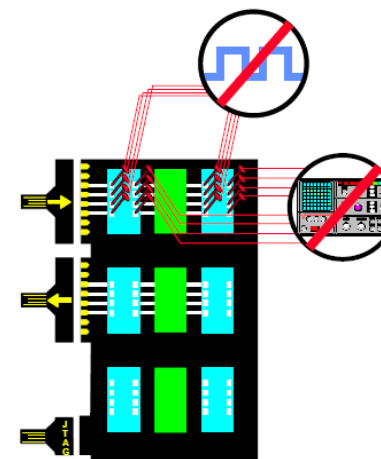
◆ ASP-Addressable Scan Port Device

JTAG in Real Chips - FPGA



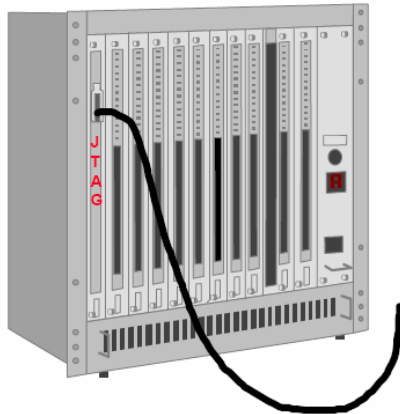
- Scan access to chips, boards, systems for:
 - Design verification/debug
 - Manufacturing test
 - Hardware/software integration
 - Field test/diagnostics
- Access built-in self-test (BIST)
- Access on-chip/in-circuit emulation (ONCE/ICE)
- Access in-system programming (ISP) of PLDs/EEPROMs
- Let your imagination run wild!!!

JTAG in Real Chips - FPGA



- Provides control and observation of system under test without need for physical access
 - Ease of set-up for test
 - Can be used in standard system configuration (no need for card extenders, etc.)
 - Can be used in environmental chambers
 - Can access on-chip emulation for software/debug
 - Can access ISP for code download/offload/ changes

System Configuration Maintenance



- Provides low-level test access within configured systems for:
 - In-house system integration
 - Fielded-system test and diagnostics
 - Built-in self-test
 - In-field upgradability via ISP, etc.
 - Remote field test, diagnostic and upgrade

IEEE Standard 1149.1



- IEEE Std 1149.1-1990 (Includes IEEE Std 1149.1a-1993), IEEE Standard Test Access Port and Boundary-Scan Architecture, ISBN 1-55937-350-4, IEEE order number SH16626.
 - *The official document which specifies the international standard for a test access port and boundary-scan architecture. Informally known as the JTAG standard, it was officially ratified by the IEEE in February 1990. Since, it has been supplemented twice. The first supplement, ratified in June 1993, is included in the referenced document. The second supplement is currently a separate document, as referenced below.*
- IEEE Std 1149.1b-1994, Supplement to IEEE Std 1149.1-1990, ISBN 1-55937-497-7, IEEE order number SH94256.
 - *The official document which specifies the international standard for a boundary-scan description language. This supplement to IEEE Std 1149.1-1990 was ratified in September 1994.*

References



- The Test Access Port and Boundary-Scan Architecture, Colin M. Maunder, Rodham E. Tulloss, ed., IEEE CS Press, ISBN 0-8186-9070-4.
 - *Edited by two principal chairs of the IEEE 1149.1 working group, this Computer Society tutorial compiles several of the seminal papers on boundary-scan along with several invited papers on various topics including applications, implementation, and others. It will primarily be of interest to the design and/or test engineer.*
- The Boundary-Scan Handbook, Kenneth P. Parker, Kluwer Academic Publishers, ISBN 0-7923-9270-1.
 - *Authored by the principal force behind the Boundary-Scan Description Language (BSDL) and an IEEE 1149.1 working group principal as well as a long time manufacturing and design-for-test expert, this is truly considered THE indispensable handbook on boundary scan for the design and/or test engineer.*
- Boundary-Scan Test - A Practical Approach, Harry Bleeker, Peter van den Eijnden, Frans de Jong, Kluwer Academic Publishers, ISBN 0-792-9296-5.
 - *Authored by several JTAG and IEEE 1149.1 working group principals, this book is a ready reference to boundary-scan technology, its benefits, and considerations for design and test managers and engineers.*

Abbreviations/Acronyms



ASIC	Application-Specific Integrated Circuit	IEEE	Institute of Electrical & Electronics Engineers IR
ASP	Addressable Scan Port		Instruction Register
ATE	Automatic Test Equipment	ISP	In-System Programming
ATPG	Automatic Test Pattern Generation	JTAG	Joint Test Action Group
BIST	Built-In Self-Test	MCM	Multi-Chip Module
B/S	Boundary-Scan	Mfg	Manufacturing
BSC	Boundary-Scan Cell	PCB	Printed Circuit Board
BSDL	Boundary-Scan Description Language	PLD	Programmable Logic Device
BSR	Boundary-Scan Register	PRPG	Pseudo-Random Pattern Generation
BST	Boundary-Scan Test	PSA	Parallel Signature Analysis
CAE	Computer-Aided Engineering	PWB	Printed Wiring Board
DFT	Design-for-Test	SPL	Scan Path Linker
DR	Data Register	SVF	Serial Vector Format
DSP	Digital Signal Processing/Processor	TAP	Test Access Port
EDA	Electronic Design Automation	TBC	Test Bus Controller
eTBC	Embedded Test Bus Controller	TCK	Test Clock
FPGA	Field-Programmable Gate Array	TDI	Test Data Input
HSDL	Hierarchical Scan Description Language	TDO	Test Data Output
ICE	In-Circuit Emulation	TMS	Test Mode Select
ICT	In-Circuit Test	TRST	Test Reset
		UUT	Unit Under Test